

Domain	Domestic hot water
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code	service	1						
DHW-1a	Control of DHW storage charging (with direct electric heating or integrated electric heat pump)	Service group: Control DHW production facilities						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Automatic control on / off	0	0	0	0	0	0	0
level 1	Automatic control on / off and scheduled charging enable	1	1	0	1	0	0	0
level 2	Automatic control on / off and scheduled charging enable and multi-sensor storage management	2	2	0	2	0	0	0
level 3	Automatic charging control based on local availability of renewables or information from electricity grid (DR, DSM)	2	3	0	2	0	0	0
level 4	0							
Information sources								
Standard?		EN 15232						

code	service	1						
DHW-1b	Control of DHW storage charging (using hot water generation)	Service group: Control DHW production facilities						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Automatic control on / off	0	0	0	0	0	0	0
level 1	Automatic control on / off and scheduled charging enable	1	1	0	1	0	0	0
level 2	Automatic on/off control, scheduled charging enable and demand-based supply temperature control or multi-sensor storage management	2	2	0	2	0	0	0
level 3	DHW production system capable of automatic charging control based on external signals (e.g. from district heating grid)	2	3	0	2	0	0	0
level 4	0							
Information sources								
		Standard?	EN 15232					

code	service	0	
DHW-1d	Control of DHW storage charging (with solar collector and supplementary heat generation)		Service group: Control DHW production facilities

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Manual selected control of solar energy or heat generation	0	0	0	0	0	0	0
level 1	Automatic control of solar storage charge (Prio. 1) and supplementary storage charge	1	0	0	1	0	0	0
level 2	Automatic control of solar storage charge (Prio. 1) and supplementary storage charge and demand-oriented supply or multi-sensor storage management	2	1	0	2	0	0	0
level 3	Automatic control of solar storage charge (Prio. 1) and supplementary storage charge, demand-oriented supply and return temperature control and multi-sensor storage management	3	2	0	2	0	0	0
level 4	a)							
Information sources								
Standard?		EN 15232						
code	service	✗0						
DHW-2b	Sequencing in case of different DHW generators	Service group: Control DHW production facilities						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Priorities only based on running time	0	0	0	0	0	0	0
level 1	Control according to fixed priority list: e.g. based on rated energy efficiency	1	0	0	0	0	0	0
level 2	Control according to dynamic priority list (based on current energy efficiency, carbon emissions and capacity of generators, e.g. solar, geothermal heat, cogeneration plant, fossil fuels)	2	1	0	0	0	0	0
level 3	Control according to dynamic priority list (based on current AND predicted load, energy efficiency, carbon emissions and capacity of generators)	3	2	0	0	0	0	0
level 4	Control according to dynamic priority list (based on current AND predicted load, energy efficiency, carbon emissions, capacity of generators AND external signals from grid)	3	3	0	0	0	0	0
Information sources								
Standard?								
code	service	✓1						
DHW-3	Report information regarding domestic hot water performance	Service group: Information to occupants and facility managers						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Indication of actual values (e.g. temperatures, submetering energy usage)	1	0	0	0	0	1	1
level 2	Actual values and historical data	1	0	0	0	0	1	2

	level 3	Performance evaluation including forecasting and/or benchmarking	1	0	0	0	0	1	3
	level 4	Performance evaluation including forecasting and/or benchmarking; also including predictive management and fault detection	1	0	0	1	0	2	3
	Information sources								
	Standard?								